Exercise Sheet 4

Model Testing and Validation

Testing Concepts

Exercise 4.1: Traditional Testing vs. ML Model Testing

Name at least four differences between testing traditional software and testing a machine learning model.

Exercise 4.2: Test Oracles *optional*

In traditional software testing, a *test oracle* determines whether a test has passed or failed.

1. What plays the role of the test oracle in ML model testing?
2. Why is defining a good test oracle particularly difficult for perception tasks?

Empirical Risk Minimization

Exercise 4.3: From Risk to Metrics *optional*

Consider binary pedestrian detection in autonomous driving.

1. Write the empirical risk minimization (ERM) objective using cross-entropy loss. Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$ be the training dataset and ℓ_{CE} the cross-entropy loss. What is $\hat{R}(\theta)$?
2. Why do we optimize cross-entropy loss instead of directly optimizing the metric we care about (e.g., recall on pedestrians)?
3. If a model has low training loss (high accuracy) but low recall on the pedestrian class, what might explain this? (Hint: think about class imbalance and what accuracy actually measures.)
4. How would you detect this problem in practice? What metrics would you compute?

Distribution Shift and Data Quality

Exercise 4.4: Distribution Shift Types

Consider the following three scenarios involving the CARLA autonomous driving model (trained on sunny daytime data):

1. The model is deployed during winter, when roads are often wet and the sun is at a low angle, creating glare in camera images.
2. A new city zone is added to the test route where $60\,\%$ of road users are cyclists - a class that was rare ($< 5\,\%$) in the original training data.
3. The city replaces all old traffic light housings with a new, slimmer design that the model has never seen.

For each scenario:

a) Identify which type of distribution shift applies (covariate shift, label shift, or concept shift). Justify your answer.

b) Explain what effect you would expect on model performance.

c) Suggest one mitigation strategy.

Exercise 4.5: ODD Coverage with k-Projection Coverage

You defined an ODD in Exercise Sheet 2. Now assess how well your *test set* covers this ODD using k-projection coverage.

1. Describe in your own words what k-projection coverage measures and why it is a useful metric for ODD coverage.
2. Download the implementation from `https://github.com/kkirchheim/odd-coverage` and compute the k-projection coverage of your test set for $k \in \{1, 2, 3\}$.
3. How does coverage change with increasing k? What does this tell you about the adequacy of your test set?

Practical: Testing the CARLA Model

Exercise 4.6: Designing a Test Suite from Safety Constraints

Refer back to Exercise 2.7 in Sheet 2, where you derived safety constraints from Unsafe Control Actions (UCAs). For each **model-level constraint**, design **one test case** using the following structure:

Constraint ID	**Constraint**	**Test input description**	**Expected output**	**Pass criterion**

Hint: Include the constraint ID (e.g., SC-1, SC-3, SC-4) in the first column. This links your test design explicitly to the safety constraints you identified.

Exercise 4.7: Per-Class Evaluation

Evaluate each of your three models on the provided test set.

1. Compute and report precision, recall, and F1-score for each model.
2. Plot a confusion matrix for each model.
3. Which model has the lowest recall? Is this the model you expected to be hardest based on your hazard analysis?

4. Based on a safety argument: what minimum recall would you require for the *pedestrian* model before considering deployment? Justify your threshold.